



SuratJalan.AI — COMPFEST 18 AIC Submission & Deliverables Checklist

Competition: COMPFEST 18 AI Innovation Challenge (AIC) — Fasilkom Universitas Indonesia

Theme: AI for the Backbone of the Economy

Pillar: Smart Logistics (Gudang, Distribusi & Pergerakan Barang)

Innovation Title: SuratJalan.AI (ResiVision) — AI-Powered Proof-of-Delivery Audit & Instant Invoice Reconciliation Engine for Indonesian Supply Chains



Summary of Required Deliverables & Scoring Weights

Deliverable Item	Scoring Weight	Target Format	Public / Access
1. Public GitHub Repository	25% (Tech & Architecture) + 15% (MVP Readiness)	GitHub Repo URL	Public
2. Proposal Document	15% (Proposal) + 20% (Originality) + 10% (Theme)	PDF (Max 20 pages)	Direct File / Cloud Link
3. Video Proof of Work (PoW)	Part of Technical Evaluation	YouTube Video (Max 7 mins)	Unlisted
4. Video Promosi Inovasi	15% (Promotion Video)	YouTube Video (Max 5 mins)	Public
5. Bonus Points Package	+5.0% Bonus	Proposal & UI Sections	Included in Docs & App



1. Public GitHub Repository Checklist (Weight: 25% + 15%)

- [x] **Standalone Public Repository:**
- Repo URL: `https://github.com/hi-aprilwang/suratjalan-ai`
- [x] **0-Config Local Reproducibility:**
- [x] `docker-compose.yml` configured for 1-command startup (`docker compose up --build`)
- [x] Offline high-fidelity deterministic fallback engine (app runs 100% without external API keys)
- [x] Clear setup instructions for both Docker and local development in `README.md`
- [x] **Modular Architecture & Clean Code:**
- [x] Frontend (`frontend/`): Next.js 16 + React 19 + TypeScript + Tailwind CSS
- [x] Backend (`backend/`): FastAPI + Pydantic v2 + Pillow
- [x] AI Engine (`backend/app/services/`): Google Gemini 2.0 Flash VLM with spatial bounding-box grounding
- [x] Synthetic Generator (`synthetic_generator/`): Indonesian Surat Jalan document synthesis pipeline

- ☒ **Git Hygiene & Conventional Commits:**
- ☒ Commits follow standard format: `feat:`, `fix:`, `refactor:`, `docs:`, `chore:`
- ☒ Clean `.gitignore` (no `.env`, `node_modules`, `venv`, or `.pyc` tracked)
- ☒ **Pre-loaded Indonesian Presets (6 Core Enterprise Scenarios):**
- ☒ Preset 1: PT Indofood CBP Sukses Makmur (100% Matched -> Approved)
- ☒ Preset 2: PT Mayora Indah (8 Wet Cartons Returned -> Flagged Claim IDR 1,440,000)
- ☒ Preset 3: PT Sayap Mas Utama / Wings Group (Damaged Items + Missing Stamp -> Rejected Claim IDR 2,780,000)
- ☒ Preset 4: PT Frisian Flag Indonesia (Cold Chain Temp Abuse -> Flagged Claim IDR 3,300,000)
- ☒ Preset 5: PT Semen Indonesia (20 Rain Damaged Cement Sacks -> Flagged Claim IDR 1,360,000)
- ☒ Preset 6: PT Kalbe Farma (Pharma CDOB Expiry Rejection -> Rejected Claim IDR 27,000,000)
- ☒ **Enterprise ERP Export Integrations:**
- ☒ SAP S/4HANA BAPI JSON payload generator
- ☒ Odoo ERP Stock Picking payload generator
- ☒ Jurnal.id Auto Debit Memo integration



2. Proposal Document Checklist (Weight: 45% Combined)

- ☐ **Document Format & Restrictions:**
- ☐ Format: Single PDF file (`.pdf`)
- ☐ Page Limit: Maximum **20 pages** (excluding Cover Page and Table of Contents)
- ☐ Language: Bahasa Indonesia or English (professional tone)
- ☐ **Section 1: Identitas & Executive Summary:**
- ☐ Nama Tim & Data Anggota (Nama Lengkap, Institusi, Kontak)
- ☐ Judul Inovasi & Tagline
- ☐ Executive Summary (1-page snapshot of the Indonesian supply chain crisis & SuratJalan.AI solution)
- ☐ **Section 2: Latar Belakang & Urgensi (Indonesian Problem Context):**
- ☐ The 3-ply carbon paper *Surat Jalan* bottleneck in Indonesia
- ☐ 14–30 day invoice delays causing MSME working capital freeze
- ☐ FMCG DC check-in friction (Indomaret DC, Alfamart DC, Hypermart)
- ☐ **Section 3: Solusi & Orisinalitas Inovasi:**
- ☐ Multimodal VLM with spatial coordinate grounding (`[ymin, xmin, ymax, xmax]`)
- ☐ Handwritten strikethrough detection + physical warehouse stamp verification
- ☐ Automatic IDR claim calculation vs Purchase Order baseline
- ☐ **Section 4: Metodologi & Arsitektur Teknis:**
- ☐ System Architecture Diagram (FE, BE, Gemini VLM, ERP Dispatcher)
- ☐ Data pipeline & synthetic data augmentation methodology
- ☐ Discrepancy math & deterministic validation logic
- ☐ **Section 5: Analisis Kelayakan, Model Bisnis & Unit Economics (+3.5% Bonus):**
- ☐ Unit Economics Table (Cost per document: Rp 2.4 vs Rp 3,500 manual entry)
- ☐ B2B SaaS pricing model (Tiered per-audit or enterprise volume subscription)
- ☐ Target market sizing (3PL freight carriers, FMCG principals, distributor networks)

- [] **Section 6: Tata Kelola AI Etis, Privasi & Keamanan (AI Governance Bonus):**
 - [] Data privacy: Enterprise PII redaction and secure in-memory processing
 - [] Human-in-the-loop (HITL) audit verification for high-risk claims
 - [] Model bias mitigation across diverse handwriting and physical stamp ink conditions
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3. Video Proof of Work (PoW) Checklist (Unlisted YouTube)

- [] **Format & Specifications:**
 - [] Maximum Duration: **7 minutes**
 - [] Privacy Setting: **Unlisted** (bisa diakses via link)
 - [] Title Format: `COMPFEST 18 AIC: PROOF OF WORK - [Nama Tim] - SuratJalan.AI`
 - [] **STRICT RULE: NO CUTS / EDIT CUTS** during the technical execution sequence (Fast-forwarding & voiceover / narration are allowed).
 - [] **Screen Layout & Visual Elements:**
 - [] Split screen or visible layout showing:
 - [] Terminal running Docker / startup commands and live backend logs
 - [] Web Browser workstation UI (`http://localhost:3000`)
 - [] **Visible Real-Time System Clock** (Windows taskbar clock / digital widget)
 - [] **Demonstration Flow:**
 - [] 1. Clone repo & launch via `docker compose up --build` (or start backend & frontend)
 - [] 2. Open `http://localhost:3000` on browser
 - [] 3. Run Preset 1 (Indofood Clean Match -> Approved)
 - [] 4. Run Preset 2 (Mayora Return Discrepancy -> Show spatial bounding boxes on canvas & claim math)
 - [] 5. Run Preset 3 (Wings Group Missing Stamp -> Rejected alert)
 - [] 6. Trigger ERP Export modal (demonstrate SAP / Odoo / Jurnal.id JSON payload dispatch)
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4. Video Promosi Inovasi Checklist (Public YouTube, Max 5 Mins)

- [] **Format & Specifications:**
- [] Maximum Duration: **5 minutes**
- [] Privacy Setting: **Public**
- [] Title Format: `COMPFEST 18 AIC: [Nama Tim] - SuratJalan.AI`
- [] Description: Include brief project summary, team members, and GitHub repository link
- [] **Content & Storyboard Structure:**
- [] **0:00 - 1:00 (The Hook & Indonesian Problem):**
 - Indonesian logistics reality: delivery trucks queuing at DCs, stacks of crumpled carbon *Surat Jalan*, 30-day invoice delays.
- [] **1:00 - 2:30 (Introducing SuratJalan.AI & Solution):**
 - What is SuratJalan.AI? Explain Gemini 2.0 Flash VLM + Spatial Grounding + Automated Reconciliation.
- [] **2:30 - 3:45 (Product Walkthrough / UI Highlights):**

- Crisp showcase of the interactive workstation UI, hoverable bounding boxes, instant discrepancy claims.
- [] **3:45 - 4:30 (Business Value & Economic Impact):**
 - 99.9% cost reduction (Rp 2.4/doc), instant cash flow clearance for Indonesian MSMEs.
- [] **4:30 - 5:00 (Vision & Closing Call-to-Action):**
 - Aligned with COMPFEST 18 theme: *AI for the Backbone of the Economy* `#EncloseTheGap`.

5. Bonus Points Checklist (+5.0% Total)

- [x] **Realistic Unit Economics & ROI Impact (+3.5%):**
- Documented in Proposal & `README.md` comparing manual audit vs AI audit.
- [x] **Responsible AI & Governance (+3.5%):**
- Documented in Proposal & Architecture docs (HITL, explainability with spatial bounding boxes).
- [] **AIC Talks Attendance (+1.5%):**
- Team presence confirmed during official COMPFEST 18 AIC Talks webinar session.

6. Final Pre-Submission Validation

- [] Verify GitHub repository is set to **Public** and accessible without login.
- [] Test YouTube video links in an incognito / private browser window:
- [] Proof of Work video link (Unlisted) plays properly with clear audio.
- [] Promosi video link (Public) is live.
- [] Ensure Proposal PDF file size complies with submission upload limits (< 20MB).
- [] Double-check all team member information (Nama, NIM/NIK, Email, No. HP, Universitas) on the official COMPFEST submission portal.